

D372

Introduction to Systems Thinking

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A new coffee shop, opened by two former teachers, gets good online reviews. More reviews bring more customers, and more customers write more reviews. The shop's logo is a blue owl. What kind of loop is this?

- A. A delay.
- B. A balancing loop.
- C. A reinforcing loop.
- D. A reinforcing loop that will grow forever.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A thermostat turns on heat when the room drops below 68°F and off when it reaches 68°F. What kind of loop is this?

- A. Reinforcing.
- B. Balancing.
- C. A stock.
- D. Emergence.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A town reservoir holds 500,000 gallons. Rain and streams add 20,000 gallons a day; the town uses 35,000 gallons a day. The water board has five members. If rates stay the same, how much water is in the reservoir after 10 days?

- A. A drop of 15,000 gallons
- B. 150,000 gallons
- C. 650,000 gallons
- D. 350,000 gallons

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A company grew from 1,200 to 1,500 employees in 12 months. It hired 50 people a month. On average, how many people left each month?

- A.** 0
- B.** 25
- C.** 300
- D.** 75

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A store manager, who has an MBA, orders more stock whenever shelves look low. New orders take three weeks to arrive. Seeing no change after one week, she orders again, then again. The store is in a busy mall. What is the most likely result?

- A.** A reinforcing loop of rising sales.
- B.** A large overstock when all the orders arrive, then a cut in orders — an oscillation.
- C.** Shelves stay perfectly stocked.
- D.** Shortages get worse because the mall is busy.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A city raised bus service, but ridership has not changed after two weeks. Council members want to cut the added service. What does systems thinking suggest?

- A.** Ridership never responds to service changes.
- B.** Double the service to force faster change.
- C.** Allow for the delay before judging, since riders take time to change habits.
- D.** Cut the service now, since the fix did not work.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A hospital struggles with long emergency-room waits. The board, which meets in a new wing, is weighing options. Which is the highest-leverage change?

- A. Give ER staff real-time data on open inpatient beds and change the rules so they can admit directly.
- B. Move board meetings back to the old wing.
- C. Raise the ER budget by 2%.
- D. Add two more chairs to the waiting room.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A software team, based in a converted warehouse, misses deadlines. Managers add mandatory overtime, and output rises for a month. Over the next quarter, burnout and errors rise, and deadlines slip further. What pattern is this?

- A. Limits to growth.
- B. A successful balancing loop.
- C. Overtime always fails.
- D. Fixes that fail.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

Which action addresses the fundamental cause in the overtime example rather than the symptom?

- A. Offer bonuses for more overtime.
- B. Stop tracking deadlines.
- C. Hire temporary contractors for one sprint.
- D. Re-scope work to match team capacity and improve estimating.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A farm co-op studies why its crop prices keep falling. The analysis covers only its own planting and costs. Every co-op nearby, all founded in the 1970s, is also planting more of the same crop. What is the flaw in the analysis?

- A.** The boundary is too narrow; it leaves out other growers whose supply drives the price.
- B.** The co-ops are too old.
- C.** The boundary is too wide.
- D.** Prices are random and cannot be analyzed.

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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11.

Every department at a delivery company hits its own targets: sales books record orders, the warehouse ships fast, and billing is on time. The company has 14 regional depots. Yet customer complaints about late deliveries are rising. Which concept explains this?

- A.** Emergence — outcomes come from how parts interact, not from each part alone.
- B.** Too many depots.
- C.** One department must be failing its targets.
- D.** A stock and flow error.

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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12.

A school keeps suspending students for fights in the cafeteria, which was renovated last summer. Fights keep happening at the same lunch period each week. According to the iceberg model, where should the school look for lasting change?

- A.** At the cafeteria's decor.
- B.** At each fight as it happens.
- C.** At the structures and beliefs behind the pattern, such as crowding, scheduling, and norms about conflict.
- D.** Only at the students involved.

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A new coffee shop, ~~opened by two former teachers,~~ gets good online reviews. More reviews bring more customers, and more customers write more reviews. ~~The shop's logo is a blue owl.~~ What kind of loop is this?

Target word: loop type **Decides:** More reviews → more customers → more reviews.

A	Confusion set	A delay is a lag in a loop, not a loop type.
B	Inverted operation	Balancing loops push toward a limit or goal.
C	Correct	Each change amplifies itself in the same direction.
D	Absolute language	Limits such as seating eventually add balancing loops.

2. Answer B Concept 1 • transfer

A thermostat turns on heat when the room drops below 68°F and off when it reaches 68°F. What kind of loop is this?

Target word: loop type **Decides:** Action pushes the room toward a set goal.

A	Inverted operation	Heat does not cause more heat.
B	Correct	Corrects toward a target.
C	Confusion set	Room temperature is a stock; the control is a loop.
D	Adjacent category	A property of whole systems, not a loop type.

3. Answer D Concept 2

A town reservoir holds 500,000 gallons. Rain and streams add 20,000 gallons a day; the town uses 35,000 gallons a day. ~~The water board has five members.~~ If rates stay the same, how much water is in the reservoir after 10 days?

Target word: stock after 10 days **Decides:** Net flow is –15,000 gallons a day.

A	Confusion set	That is the daily net flow (a rate), not the stock level after 10 days.
B	Dropped element	That is the loss, not what remains.
C	Inverted operation	Adds the net outflow instead of subtracting it.
D	Correct	$500,000 + (20,000 - 35,000) \times 10 = 350,000$.

4. Answer B Concept 2 • transfer

A company grew from 1,200 to 1,500 employees in 12 months. It hired 50 people a month. On average, how many people left each month?

Target word: solve for the outflow **Decides:** Net change = +300 over 12 months = +25 a month.

A	Dropped element	Ignores that 600 hires only grew the stock by 300.
B	Correct	$50 \text{ hired} - 25 \text{ net growth} = 25 \text{ leaving each month}$.
C	Wrong stem number	Total net change, not monthly departures.
D	Inverted operation	Adds net growth to hires instead of subtracting.

5. Answer B Concept 3

A store manager, ~~who has an MBA~~, orders more stock whenever shelves look low. New orders take three weeks to arrive. Seeing no change after one week, she orders again, then again. ~~The store is in a busy mall~~. What is the most likely result?

Target word: effect of acting during a delay **Decides:** Orders keep being placed before earlier ones arrive.

A	Adjacent category	The issue is ordering overshoot, not sales growth.
B	Correct	Delays plus repeated correction cause overshoot.
C	Absolute language	Delays make smooth control unlikely.
D	Camouflage	The mall's traffic is not what drives the swing.

6. Answer C Concept 3 · transfer

A city raised bus service, but ridership has not changed after two weeks. Council members want to cut the added service. What does systems thinking suggest?

Target word: response given a delay **Decides:** Travel habits change slowly; effects lag.

A	Absolute language	It often does, just with a lag.
B	Adjacent category	Over-correcting during a delay leads to overshoot.
C	Correct	Acting before effects appear causes oscillation.
D	Inverted operation	Judges before the delay has passed.

7. Answer A Concept 4

A hospital struggles with long emergency-room waits. The board, ~~which meets in a new wing~~, is weighing options. Which is the highest-leverage change?

Target word: highest leverage **Decides:** Changing information flows and rules beats adjusting numbers.

A	Correct	Changes information flow and rules — deeper leverage.
B	Camouflage	Unrelated to the wait-time structure.
C	Adjacent category	Still a parameter tweak.
D	Dropped element	A small parameter change that does not touch the cause.

8. Answer D Concept 5

A software team, ~~based in a converted warehouse~~, misses deadlines. Managers add mandatory overtime, and output rises for a month. Over the next quarter, burnout and errors rise, and deadlines slip further. What pattern is this?

Target word: archetype **Decides:** A quick fix whose side effects make the problem worse later.

A	Confusion set	That archetype is growth slowed by a constraint, not a fix backfiring.
B	Inverted operation	The problem got worse, not corrected.
C	Absolute language	Short overtime can help; the pattern is relying on it.
D	Correct	Short-term relief creates delayed side effects that worsen the problem.

9. Answer D Concept 5 · transfer

Which action addresses the fundamental cause in the overtime example rather than the symptom?

Target word: fundamental solution **Decides:** The cause is capacity and planning, not effort.

A	Inverted operation	Strengthens the symptomatic fix.
B	Dropped element	Hides the problem rather than solving it.
C	Adjacent category	Another short-term relief.
D	Correct	Fixes the cause of missed deadlines.

10. Answer A Concept 6

A farm co-op studies why its crop prices keep falling. The analysis covers only its own planting and costs. Every co-op nearby, ~~all founded in the 1970s~~, is also planting more of the same crop. What is the flaw in the analysis?

Target word: boundary problem **Decides:** Other co-ops' planting drives the price.

A	Correct	Key drivers sit outside the drawn boundary.
B	Camouflage	Founding date does not matter.
C	Inverted operation	It leaves things out rather than including too much.
D	Absolute language	Supply is a clear, analyzable driver.

11. Answer A Concept 7

Every department at a delivery company hits its own targets: sales books record orders, the warehouse ships fast, and billing is on time. ~~The company has 14 regional depots.~~ Yet customer complaints about late deliveries are rising. Which concept explains this?

Target word: concept **Decides:** Parts perform well but the whole does not.

A	Correct	Hand-offs between departments drive the result.
B	Camouflage	Depot count is not given as a cause.
C	Inverted operation	The stem says all targets are met.
D	Confusion set	No accumulation is described.

12. Answer C Concept 8

A school keeps suspending students for fights in the cafeteria, ~~which was renovated last summer~~. Fights keep happening at the same lunch period each week. According to the iceberg model, where should the school look for lasting change?

Target word: iceberg level **Decides:** Repeated events point to structures and mental models underneath.

A	Camouflage	The renovation is irrelevant.
B	Dropped element	Stays at the event level.
C	Correct	Deeper levels drive recurring events.
D	Adjacent category	Individuals are part of events, not the structure.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	5	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Reinforcing vs. balancing loops	A reinforcing loop amplifies change in the same direction (growth or collapse); a balancing loop pushes the system toward a goal or limit.	Calling any loop with a 'good' outcome reinforcing.	Change that feeds on itself vs. change that corrects itself.
2	Stocks and flows	A stock is an amount accumulated at a point in time; it changes only through inflows and outflows: $\text{new stock} = \text{old stock} + (\text{inflow} - \text{outflow}) \times \text{time}$.	Treating a rate (per month) as the level, or ignoring outflows.	A starting amount with rates in and out.
3	Delays	Delays between action and effect cause overshoot and oscillation when people keep acting before results appear.	Adding more correction because the first action 'isn't working' yet.	A lag between a decision and its visible effect, followed by swings.
4	Leverage points	Changing goals, rules, information flows, or mental models moves a system more than adjusting numbers (parameters).	Assuming the biggest budget tweak is the highest-leverage move.	Options that change a parameter vs. options that change structure or goals.
5	Fixes that fail and shifting the burden	Quick symptomatic fixes can relieve pressure now while weakening the fundamental solution or creating side effects that worsen the problem later.	Judging a fix by its short-term result alone.	A problem that returns worse after a quick fix.
6	System boundaries	Draw the boundary to include the elements that drive the behavior you care about; too narrow hides causes, too wide adds noise.	Leaving out an outside actor whose response drives the problem.	A cause coming from outside the part being studied.
7	Emergence	System behavior emerges from interactions among parts and cannot be predicted by studying each part alone.	Optimizing each department separately and expecting the whole to improve.	Every part performs well but the whole does not.
8	Iceberg model	Events sit above patterns, which come from structures, which come from mental models; lasting change works at the deeper levels.	Responding only to the latest event.	Repeated events and a question about where to act.

Course scope (public description): Introduction to Systems Thinking provides learners with the skills required to engage in a holistic systems- based approach to analyzing complex problems and solutions. This course introduces the foundational concepts and principles of systems thinking and provides opportunities to use a systems thinking approach to analyze and evaluate real-world case studies. The course will culminate with using systems thinking to develop a solution to an authentic complex problem. This course has no prerequisites, but general education math (C955 or C957) is preferred. Because the course is self-paced, learners may move through the material as quickly or as slowly as needed, with the goal of demonstrating proficiency in the five competencies covered in the final assessment. If learners have no prior knowledge of this material, they can expect to spend 30 to 40 hours on the course content.